



Frames, Objects, and Scripts – OSP-5 Notes

1. Frames

Definition:

A **Frame** is a **data structure** for representing a stereotyped situation or object, similar to a record or a class in object-oriented programming. It contains **slots (attributes)** and **slot values** (data or procedures).

Structure:

- **Frame Name:** The concept (e.g., “Car”)
- **Slots:** Properties or features (e.g., “Color”, “Wheels”)
- **Slot Values:** The actual data or default values

Example:

yaml

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Frame: Car

- Wheels: 4
- Engine: Present
- Color: Variable
- FuelType: Petrol/Diesel/Electric

Features:



- **Inheritance:** Sub-frames can inherit attributes from parent frames.
- **Default Reasoning:** Slot values can have default assumptions.
- **Procedural Attachments:** Functions can be attached to slots.

Application:

Used in **expert systems**, **robotics**, **scene understanding**, and **NLP**.

2. Objects

Definition:

In AI, an **Object** is an **instance** of a class (or frame) that contains both **data (attributes)** and **methods (behaviors)**, following the **object-oriented paradigm**.

Structure:

- **Class:** Template for a group of objects.
- **Object:** A specific instance with real values.

Example:

python

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Class: Dog

- Attributes: Breed, Age, Color
- Method: Bark()

Object: MyDog

- Breed: Labrador
- Age: 3
- Color: Brown



Features:

- **Encapsulation:** Combines data and behavior.
- **Reusability:** Code and logic can be reused across objects.
- **Modularity:** Easy to maintain and expand.

Application:

Widely used in **AI programming, simulation, and knowledge systems.**

3. Scripts

Definition:

A **Script** is a structured representation of a **sequence of events** in a typical situation. It models the **temporal flow** of actions and expected behaviors.

Purpose:

To help AI systems understand **common human activities** and respond appropriately.

Structure:

- **Entry Conditions:** What must be true before the script starts
- **Scenes:** Sequence of events or actions
- **Roles:** Participants in the scenario
- **Results:** What is expected after the script completes



Example:

Script: Restaurant Visit

- Entry: Hungry person goes to restaurant
- Scenes: Enter → Order → Eat → Pay → Leave
- Roles: Customer, Waiter, Chef, Cashier
- Result: Customer is full, restaurant is paid

Features:

- Captures **procedural knowledge**
- Deals with **expectations** and **defaults**
- Helps with **language understanding** and **story comprehension**

Application:

Used in **natural language processing**, **story understanding**, **dialogue systems**, and **AI reasoning**.

Summary Table

Concept	Structure	Key Feature	Application Area
Frame	Slots + Values	Represents knowledge in attributes	Expert systems, robotics
Object	Attributes + Methods	Encapsulation of data + behavior	Programming, AI models
Script	Scenes + Roles	Sequence of events (procedural)	NLP, story/dialogue systems